We begin by defining the concept of Riemann integrability of real-valued functions defined on a closed bounded interval of $\mathbb{R}$, using the Riemann sums familiar to the reader from calculus. This method has the advantage that it extends immediately to the case of functions whose values are complex numbers, or vectors in the space $\mathbb{R}^n$. In Section 7.2, we will establish the Riemann integrability of several important classes of functions: step functions, continuous functions, and monotone functions. However, we will also see that there are functions that are *not* Riemann integrable. The Fundamental Theorem of Calculus is the principal result in Section 7.3. We will present it in a form that is slightly more general than is customary and does not require the function to be a derivative at every point of the interval. A number of important consequences of the Fundamental Theorem are also given. In Section 7.3 we also give a statement of the definitive Lebesgue Criterion for Riemann integrability. This famous result is usually not given in books at this level, since its proof (given in Appendix C) is somewhat complicated. However, its statement is well within the reach of students, who will also comprehend the power of this result. In Section 7.4, we discuss an alternative approach to the Riemann integral due to Gaston Darboux that uses the concepts of upper integral and lower integral. The two approaches appear to be quite different, but in fact they are shown to be equivalent. The final section presents several methods of approximating integrals, a subject that has become increasingly important during this era of high-speed computers. While the proofs of these results are not particularly difficult, we defer them to Appendix D.

An interesting history of integration theory, including a chapter on the Riemann integral, is given in the book by Hawkins cited in the References.

Section 7.1 Riemann Integral

We will follow the procedure commonly used in calculus courses and define the Riemann integral as a kind of limit of the Riemann sums as the norm of the partitions tend to 0. Since we assume that the reader is familiar—at least informally—with the integral from a calculus course, we will not provide a motivation of the integral, or discuss its interpretation as the "area under the graph," or its many applications to physics, engineering, economics, etc. Instead, we will focus on the purely mathematical aspects of the integral.

However, we first define some basic terms that will be frequently used.

Partitions and Tagged Partitions

If $I := [a, b]$ is a closed bounded interval in $\mathbb{R}$, then a **partition** of I is a finite, ordered set $\mathcal{P} := (x_0, x_1, \ldots, x_{n-1}, x_n)$ of points in I such that

$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b.$$

(See Figure 7.1.1.) The points of $\mathcal{P}$ are used to divide $I = [a, b]$ into non-overlapping subintervals

$$I_1 := [x_0, x_1], \quad I_2 := [x_1, x_2], \ldots, \quad I_n := [x_{n-1}, x_n].$$

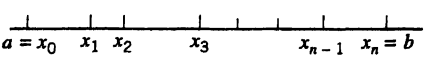


Figure 7.1.1 A partition of $[a, b]$

Often we will denote the partition $\mathcal{P}$ by the notation $\mathcal{P} = \{[x_{i-1}, x_i]\}_{i=1}^n$. We define the **norm** (or **mesh**) of $\mathcal{P}$ to be the number

$$(1) \qquad \|\mathcal{P}\| := \max\{x_1 - x_0, x_2 - x_1, \ldots, x_n - x_{n-1}\}.$$

Thus the norm of a partition is merely the length of the largest subinterval into which the partition divides $[a, b]$. Clearly, many partitions have the same norm, so the partition is *not* a function of the norm.

If a point t_i has been selected from each subinterval $I_i = [x_{i-1}, x_i]$, for $i = 1, 2, \ldots, n$, then the points are called **tags** of the subintervals I_i. A set of ordered pairs

$$\dot{\mathcal{P}} := \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$$

of subintervals and corresponding tags is called a **tagged partition** of I; see Figure 7.1.2. (The dot over the $\mathcal{P}$ indicates that a tag has been chosen for each subinterval.) The tags can be chosen in a wholly arbitrary fashion; for example, we can choose the tags to be the left endpoints, or the right endpoints, or the midpoints of the subintervals, etc. Note that an endpoint of a subinterval can be used as a tag for two consecutive subintervals. Since each tag can be chosen in infinitely many ways, each partition can be tagged in infinitely many ways. The norm of a tagged partition is defined as for an ordinary partition and does not depend on the choice of tags.

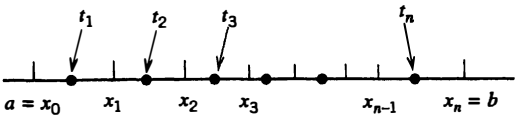


Figure 7.1.2 A tagged partition of $[a, b]$

If $\dot{\mathcal{P}}$ is the tagged partition given above, we define the **Riemann sum** of a function $f : [a, b] \to \mathbb{R}$ corresponding to $\dot{\mathcal{P}}$ to be the number

$$(2) \qquad S(f; \dot{\mathcal{P}}) := \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1}).$$

We will also use this notation when $\dot{\mathcal{P}}$ denotes a *subset* of a partition, and not the entire partition.

The reader will perceive that if the function f is positive on $[a, b]$, then the Riemann sum (2) is the sum of the areas of n rectangles whose bases are the subintervals $I_i = [x_{i-1}, x_i]$ and whose heights are $f(t_i)$. (See Figure 7.1.3.)

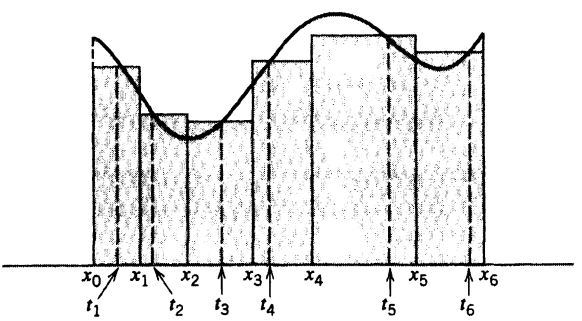


Figure 7.1.3 A Riemann sum

Definition of the Riemann Integral

We now define the Riemann integral of a function f on an interval $[a, b]$.

7.1.1 Definition A function $f : [a, b] \to \mathbb{R}$ is said to be **Riemann integrable** on $[a, b]$ if there exists a number $L \in \mathbb{R}$ such that for every $\varepsilon > 0$ there exists $\delta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ is any tagged partition of $[a, b]$ with $||\dot{\mathcal{P}}|| < \delta_\varepsilon$, then

$$\left|S(f; \dot{\mathcal{P}}) - L\right| < \varepsilon.$$

The set of all Riemann integrable functions on $[a, b]$ will be denoted by $\mathcal{R}[a, b]$.

Remark It is sometimes said that the integral L is "the limit" of the Riemann sums $S(f : \dot{\mathcal{P}})$ as the norm $||\dot{\mathcal{P}}|| \to 0$. However, since $S(f; \dot{\mathcal{P}})$ is not a function of $||\dot{\mathcal{P}}||$, this **limit** is not of the type that we have studied before.

First we will show that if $f \in \mathcal{R}[a, b]$, then the number L is uniquely determined. It will be called the **Riemann integral of** f over $[a, b]$. Instead of L, we will usually write

$$L = \int_a^b f \quad \text{or} \quad \int_a^b f(x)dx.$$

It should be understood that any letter other than x can be used in the latter expression, so long as it does not cause any ambiguity.

7.1.2 Theorem *If $f \in \mathcal{R}[a, b]$, then the value of the integral is uniquely determined.*

Proof. Assume that L' and L'' both satisfy the definition and let $\varepsilon > 0$. Then there exists $\delta'_{\varepsilon/2} > 0$ such that if $\dot{\mathcal{P}}_1$ is any tagged partition with $||\dot{\mathcal{P}}_1|| < \delta'_{\varepsilon/2}$, then

$$\left|S(f; \dot{\mathcal{P}}_1) - L'\right| < \varepsilon/2.$$

Also there exists $\delta''_{\varepsilon/2} > 0$ such that if $\dot{\mathcal{P}}_2$ is any tagged partition with $||\dot{\mathcal{P}}_2|| < \delta''_{\varepsilon/2}$, then

$$\left|S(f; \dot{\mathcal{P}}_2) - L''\right| < \varepsilon/2.$$

Now let $\delta_\varepsilon := \min\{\delta'_{\varepsilon/2}, \delta''_{\varepsilon/2}\} > 0$ and let $\dot{\mathcal{P}}$ be a tagged partition with $||\dot{\mathcal{P}}|| < \delta_\varepsilon$. Since both $||\dot{\mathcal{P}}|| < \delta'_{\varepsilon/2}$ and $||\dot{\mathcal{P}}|| < \delta''_{\varepsilon/2}$, then

$$\left|S(f; \dot{\mathcal{P}}) - L'\right| < \varepsilon/2 \qquad \text{and} \qquad \left|S(f; \dot{\mathcal{P}}) - L''\right| < \varepsilon/2,$$

whence it follows from the Triangle Inequality that

$$\begin{aligned} |L' - L''| &= \left|L' - S(f; \dot{\mathcal{P}}) + S(f; \dot{\mathcal{P}}) - L''\right| \\ &\leq \left|L' - S(f; \dot{\mathcal{P}})\right| + \left|S(f; \dot{\mathcal{P}}) - L''\right| \\ &< \varepsilon/2 + \varepsilon/2 = \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, it follows that $L' = L''$. Q.E.D.

The next result shows that changing a function at a finite number of points does not affect its integrability nor the value of its integral.

7.1.3 Theorem If g is Riemann integrable on $[a, b]$ and if $f(x) = g(x)$ except for a finite number of points in $[a, b]$, then f is Riemann integrable and $\int_a^b f = \int_a^b g$.

Proof. If we prove the assertion for the case of one exceptional point, then the extension to a finite number of points is done by a standard induction argument, which we leave to the reader.

Let c be a point in the interval and let $L = \int_a^b g$. Assume that $f(x) = g(x)$ for all $x \neq c$. For any tagged partition $\dot{\mathcal{P}}$, the terms in the two sums $S(f; \dot{\mathcal{P}})$ and $S(g; \dot{\mathcal{P}})$ are identical with the exception of at most two terms (in the case that $c = x_i = x_{i-1}$ is an endpoint). Therefore, we have

$$|S(f; \dot{\mathcal{P}}) - S(g; \dot{\mathcal{P}})| = |\Sigma(f(x_i) - g(x_i))(x_i - x_{i-1})| \leq 2(|g(c)| + |f(c)|)\,||\dot{\mathcal{P}}||.$$

Now, given $\varepsilon > 0$, we let $\delta_1 > 0$ satisfy $\delta_1 < \varepsilon/(4(|f(c)| + |g(c)|)$, and let $\delta_2 > 0$ be such that $||\dot{\mathcal{P}}|| < \delta_2$ implies $|S(g; \dot{\mathcal{P}}) - L| < \varepsilon/2$. We now let $\delta := \min\{\delta_1, \delta_2\}$. Then, if $||\dot{\mathcal{P}}|| < \delta$, we obtain

$$|S(f; \dot{\mathcal{P}}) - L| \leq |S(f; \dot{\mathcal{P}}) - S(g; \dot{\mathcal{P}})| + |S(g; \dot{\mathcal{P}}) - L| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Hence, the function f is integrable with integral L. Q.E.D.

Some Examples

If we use only the definition, in order to show that a function f is Riemann integrable we must (i) know (or guess correctly) the value L of the integral, and (ii) construct a δ_ε that will suffice for an arbitrary $\varepsilon > 0$. The determination of L is sometimes done by calculating Riemann sums and guessing what L must be. The determination of δ_ε is likely to be difficult.

In actual practice, we usually show that $f \in \mathcal{R}[a, b]$ by making use of some of the theorems that will be given later.

7.1.4 Examples **(a)** Every constant function on $[a, b]$ is in $\mathcal{R}[a, b]$.

Let $f(x) := k$ for all $x \in [a, b]$. If $\dot{\mathcal{P}} := \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ is any tagged partition of $[a, b]$, then it is clear that

$$S(f; \dot{\mathcal{P}}) = \sum_{i=1}^{n} k(x_i - x_{i-1}) = k(b - a).$$

Hence, for any $\varepsilon > 0$, we can choose $\delta_\varepsilon := 1$ so that if $||\dot{\mathcal{P}}|| < \delta_\varepsilon$, then

$$|S(f; \dot{\mathcal{P}}) - k(b - a)| = 0 < \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we conclude that $f \in \mathcal{R}[a, b]$ and $\int_a^b f = k(b - a)$.

(b) Let $g : [0, 3] \to \mathbb{R}$ be defined by $g(x) := 2$ for $0 \leq x \leq 1$, and $g(x) := 3$ for $1 < x \leq 3$. A preliminary investigation, based on the graph of g (see Figure 7.1.4), suggests that we might expect that $\int_0^3 g = 8$.

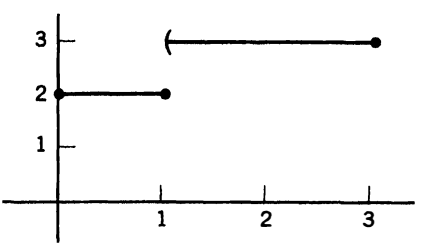


Figure 7.1.4 Graph of g

Let $\dot{\mathcal{P}}$ be a tagged partition of [0,3] with norm $< \delta$; we will show how to determine δ in order to ensure that $|S(g; \dot{\mathcal{P}}) - 8| < \varepsilon$. Let $\dot{\mathcal{P}}_1$ be the subset of $\dot{\mathcal{P}}$ having its tags in [0,1] where $g(x) = 2$, and let $\dot{\mathcal{P}}_2$ be the subset of $\dot{\mathcal{P}}$ with its tags in (1, 3] where $g(x) = 3$. It is obvious that we have

$$S(g; \dot{\mathcal{P}}) = S(g; \dot{\mathcal{P}}_1) + S(g; \dot{\mathcal{P}}_2). \tag{3}$$

If we let U_1 denote the union of the subintervals in $\dot{\mathcal{P}}_1$, then it is readily shown that

$$[0, 1 - \delta] \subset U_1 \subset [0, 1 + \delta]. \tag{4}$$

For example, to prove the first inclusion, we let $u \in [0, 1 - \delta]$. Then u lies in an interval $I_k := [x_{k-1}, x_k]$ of $\dot{\mathcal{P}}_1$, and since $||\dot{\mathcal{P}}|| < \delta$, we have $x_k - x_{k-1} < \delta$. Then $x_{k-1} \leq u \leq 1 - \delta$ implies that $x_k \leq x_{k-1} + \delta \leq (1 - \delta) + \delta \leq 1$. Thus the tag t_k in I_k satisfies $t_k \leq 1$ and therefore u belongs to a subinterval whose tag is in [0,1], that is, $u \in U_1$. This proves the first inclusion in (4), and the second inclusion can be shown in the same manner. Since $g(t_k) = 2$ for the tags of $\dot{\mathcal{P}}_1$ and since the intervals in (4) have lengths $1 - \delta$ and $1 + \delta$, respectively, it follows that

$$2(1 - \delta) \leq S(g; \dot{\mathcal{P}}_1) \leq 2(1 + \delta).$$

A similar argument shows that the union of all subintervals with tags $t_i \in (1, 3]$ contains the interval $[1 + \delta, 3]$ of length $2 - \delta$, and is contained in $[1 - \delta, 3]$ of length $2 + \delta$. Therefore,

$$3(2 - \delta) \leq S(g; \dot{\mathcal{P}}_2) \leq 3(2 + \delta).$$

Adding these inequalities and using equation (3), we have

$$8 - 5\delta \leq S(g; \dot{\mathcal{P}}) = S(g; \dot{\mathcal{P}}_1) + S(g; \dot{\mathcal{P}}_2) \leq 8 + 5\delta,$$

whence it follows that

$$|S(g; \dot{\mathcal{P}}) - 8| \leq 5\delta.$$

To have this final term $< \varepsilon$, we are led to take $\delta_\varepsilon < \varepsilon/5$.

Making such a choice (for example, if we take $\delta_\varepsilon := \varepsilon/10$), we can retrace the argument and see that $|S(g; \dot{\mathcal{P}}) - 8| < \varepsilon$ when $||\dot{\mathcal{P}}|| < \delta_\varepsilon$. Since $\varepsilon > 0$ is arbitrary, we have proved that $g \in \mathcal{R}[0, 3]$ and that $\int_0^3 g = 8$, as predicted,

(c) Let $h(x) := x$ for $x \in [0, 1]$; we will show that $h \in \mathcal{R}[0, 1]$.

We will employ a "trick" that enables us to guess the value of the integral by considering a particular choice of the tag points. Indeed, if $\{I_i\}_{i=1}^n$ is any partition of [0,1] and we choose the tag of the interval $I_i = [x_{i-1}, x_i]$ to be the midpoint $q_i := \frac{1}{2}(x_{i-1} + x_i)$, then the contribution of this term to the Riemann sum corresponding to the tagged partition $\dot{\mathcal{Q}} := \{(I_i, q_i)\}_{i=1}^n$ is

$$h(q_i)(x_i - x_{i-1}) = \frac{1}{2}(x_i + x_{i-1})(x_i - x_{i-1}) = \frac{1}{2}(x_i^2 - x_{i-1}^2).$$

If we add these terms and note that the sum telescopes, we obtain

$$S(h; \dot{\mathcal{Q}}) = \sum_{i=1}^{n} \frac{1}{2}(x_i^2 - x_{i-1}^2) = \frac{1}{2}(1^2 - 0^2) = \frac{1}{2}.$$

Now let $\dot{\mathcal{P}} := \{(I_i, t_i)\}_{i=1}^n$ be an arbitrary tagged partition of [0,1] with $||\dot{\mathcal{P}}|| < \delta$ so that $x_i - x_{i-1} < \delta$ for $i = 1, \ldots, n$. Also let $\dot{\mathcal{Q}}$ have the same partition points, but where we

choose the tag q_i to be the midpoint of the interval I_i. Since both t_i and q_i belong to this interval, we have $|t_i - q_i| < \delta$. Using the Triangle Inequality, we deduce

$$\left|S(h;\dot{\mathcal{P}}) - S(h;\ \dot{\mathcal{Q}})\right| = \left|\sum_{i=1}^{n} t_i(x_i - x_{i-1}) - \sum_{i-1}^{n} q_i(x_i - x_{i-1})\right|$$

$$\leq \sum_{i=1}^{n} |t_i - q_i|(x_i - x_{i-1}) < \delta \sum_{i=1}^{n} (x_i - x_{i-1}) = \delta(x_n - x_0) = \delta.$$

Since $S(h;\ \dot{\mathcal{Q}}) = \frac{1}{2}$, we infer that if $\dot{\mathcal{P}}$ is any tagged partition with $||\dot{\mathcal{P}}|| < \delta$, then

$$\left|S(h;\ \dot{\mathcal{P}}) - \frac{1}{2}\right| < \delta.$$

Therefore we are led to take $\delta_\varepsilon \leq \varepsilon$. If we choose $\delta_\varepsilon := \varepsilon$, we can retrace the argument to conclude that $h \in \mathcal{R}[0,\ 1]$ and $\int_0^1 h = \int_0^1 x\,dx = \frac{1}{2}$.

(d) Let $G(x) := 1/n$ for $x = 1/n$ $(n \in \mathbb{N})$, and $G(x) := 0$ elsewhere on $[0,\ 1]$.

Given $\varepsilon > 0$, the set $E := \{x : G(x) \geq \varepsilon\}$ is a finite set. (For example, if $\varepsilon = 1/10$, then $E = \{1,\ 1/2,\ 1/3, \ldots, 1/10\}$.) If n is the number of points in E, we allow for the possibility that a tag may be counted twice if it is an endpoint and let $\delta := \varepsilon/2n$. For a given tagged partition $\dot{\mathcal{P}}$ such that $||\dot{\mathcal{P}}|| < \delta$, we let $\dot{\mathcal{P}}_0$ be the subset of $\dot{\mathcal{P}}$ with all tags outside of E and let $\dot{\mathcal{P}}_1$ be the subset of $\dot{\mathcal{P}}$ with one or more tags in E. Since $G(x) < \varepsilon$ for each x outside of E and $G(x) \leq 1$ for all x in [0,1], we get

$$0 \leq S(G;\ \dot{\mathcal{P}}) = S(G;\ \dot{\mathcal{P}}_0) + S(G;\ \dot{\mathcal{P}}_1) < \varepsilon + (2n)\delta = 2\varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we conclude that G is Riemann integrable with integral equal to zero. □

Some Properties of the Integral

The difficulties involved in determining the value of the integral and of δ_ε suggest that it would be very useful to have some general theorems. The first result in this direction enables us to form certain algebraic combinations of integrable functions.

7.1.5 Theorem *Suppose that f and g are in $\mathcal{R}[a,\ b]$. Then:*

(a) If $k \in \mathbb{R}$, *the function kf is in $\mathcal{R}[a,\ b]$ and*

$$\int_a^b kf = k\int_a^b f.$$

(b) *The function $f + g$ is in $\mathcal{R}[a,\ b]$ and*

$$\int_a^b (f+g) = \int_a^b f + \int_a^b g.$$

(c) *If $f(x) \leq g(x)$ for all $x \in [a, b]$, then*

$$\int_a^b f \leq \int_a^b g.$$

Proof. If $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ is a tagged partition of $[a, b]$, then it is an easy exercise to show that

$$S(kf; \dot{\mathcal{P}}) = kS(f; \dot{\mathcal{P}}), \qquad S(f+g; \dot{\mathcal{P}}) = S(f; \dot{\mathcal{P}}) + S(g; \dot{\mathcal{P}}),$$
$$S(f; \dot{\mathcal{P}}) \leq S(g; \dot{\mathcal{P}}).$$

We leave it to the reader to show that the assertion (a) follows from the first equality. As an example, we will complete the proofs of (b) and (c).

Given $\varepsilon > 0$, we can use the argument in the proof of the Uniqueness Theorem 7.1.2 to construct a number $\delta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ is any tagged partition with $||\dot{\mathcal{P}}|| < \delta_\varepsilon$, then both

$$\left|S(f; \dot{\mathcal{P}}) - \int_a^b f\right| < \varepsilon/2 \qquad \text{and} \qquad \left|S(g; \dot{\mathcal{P}}) - \int_a^b g\right| < \varepsilon/2. \tag{5}$$

To prove (b), we note that

$$\begin{aligned}\left|S(f+g; \dot{\mathcal{P}}) - \left(\int_a^b f + \int_a^b g\right)\right| &= \left|S(f; \dot{\mathcal{P}}) + S(g; \dot{\mathcal{P}}) - \int_a^b f - \int_a^b g\right| \\ &\leq \left|S(f; \dot{\mathcal{P}}) - \int_a^b f\right| + \left|S(g; \dot{\mathcal{P}}) - \int_a^b g\right| \\ &< \varepsilon/2 + \varepsilon/2 = \varepsilon.\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, we conclude that $f + g \in \mathcal{R}[a, b]$ and that its integral is the sum of the integrals of f and g.

To prove (c), we note that the Triangle Inequality applied to (5) implies

$$\int_a^b f - \varepsilon/2 < S(f; \dot{\mathcal{P}}) \qquad \text{and} \qquad S(g; \dot{\mathcal{P}}) < \int_a^b g + \varepsilon/2.$$

If we use the fact that $S(f; \dot{\mathcal{P}}) \leq S(g; \dot{\mathcal{P}})$, we have

$$\int_a^b f \leq \int_a^b g + \varepsilon.$$

But, since $\varepsilon > 0$ is arbitrary, we conclude that $\int_a^b f \leq \int_a^b g$. Q.E.D.

Boundedness Theorem

We now show that an unbounded function cannot be Riemann integrable.

7.1.6 Theorem *If $f \in \mathcal{R}[a, b]$, then f is bounded on $[a, b]$.*

Proof. Assume that f is an unbounded function in $\mathcal{R}[a, b]$ with integral L. Then there exists $\delta > 0$ such that if $\dot{\mathcal{P}}$ is any tagged partition of $[a, b]$ with $||\dot{\mathcal{P}}|| < \delta$, then we have $\left|S(f; \dot{\mathcal{P}}) - L\right| < 1$, which implies that

$$\left|S(f; \dot{\mathcal{P}})\right| < |L| + 1. \tag{6}$$

Now let $\mathcal{Q} = \{[x_{i-1}, x_i]\}_{n=1}^n$ be a partition of $[a, b]$ with $||\mathcal{Q}|| < \delta$. Since $|f|$ is not bounded on $[a, b]$, then there exists at least one subinterval in $\mathcal{Q}$, say $[x_{k-1}, x_k]$, on which $|f|$ is not bounded—for, if $|f|$ is bounded on each subinterval $[x_{i-1}, x_i]$ by M_i, then it is bounded on $[a, b]$ by $\max\{M_1, \ldots, M_n\}$.

We will now pick tags for $\mathcal{Q}$ that will provide a contradiction to (6). We tag $\mathcal{Q}$ by $t_i := x_i$ for $i \neq k$ and we pick $t_k \in [x_{k-1}, x_k]$ such that

$$|f(t_k)(x_k - x_{k-1})| > |L| + 1 + \left|\sum_{i \neq k} f(t_i)(x_i - x_{i-1})\right|.$$

From the Triangle Inequality (in the form $|A + B| \geq |A| - |B|$), we have

$$\left|S(f; \dot{\mathcal{Q}})\right| \geq |f(t_k)(x_k - x_{k-1})| - \left|\sum_{i \neq k} f(t_i)(x_i - x_{i-1})\right| > |L| + 1,$$

which contradicts (6). Q.E.D.

We will close this section with an example of a function that is discontinuous at every rational number and is not monotone, but is Riemann integrable nevertheless.

7.1.7 Example We consider Thomae's function $h : [0, 1] \to \mathbb{R}$ defined, as in Example 5.1.6(h), by $h(x) := 0$ if $x \in [0, 1]$ is irrational, $h(0) := 1$ and by $h(x) := 1/n$ if $x \in [0, 1]$ is the rational number $x = m/n$ where $m, n \in \mathbb{N}$ have no common integer factors except 1. It was seen in 5.1.6(h) that h is continuous at every irrational number and discontinuous at every rational number in [0, 1]. See Figure 5.1.2. We will now show that $h \in \mathcal{R}[0, 1]$.

For $\varepsilon > 0$, the set $E := \{x \in [0, 1] : h(x) \geq \varepsilon/2\}$ is a finite set. (For example, if $\varepsilon/2 = 1/5$, then there are eleven values of x such that $h(x) \geq 1/5$, namely, $E = \{0, 1, 1/2, 1/3, 2/3, 1/4, 3/4, 1/5, 2/5, 3/5, 4/5\}$. (Sketch a graph.) We let n be the number of elements in E and take $\delta := \varepsilon/(4n)$. If $\dot{\mathcal{P}}$ is a given tagged partition such that $||\dot{\mathcal{P}}|| < \delta$, then we separate $\dot{\mathcal{P}}$ into two subsets. We let $\dot{\mathcal{P}}_1$ be the collection of tagged intervals in $\dot{\mathcal{P}}$ that have their tags in E, and we let $\dot{\mathcal{P}}_2$ be the subset of tagged intervals in $\dot{\mathcal{P}}$ that have their tags elsewhere in [0, 1]. Allowing for the possibility that a tag of $\dot{\mathcal{P}}_1$ may be an endpoint of adjacent intervals, we see that $\dot{\mathcal{P}}_1$ has at most $2n$ intervals and the total length of these intervals can be at most $2n\delta = \varepsilon/2$. Also, we have $0 < h(t_i) \leq 1$ for each tag t_i in $\dot{\mathcal{P}}_1$. Consequently, we have $S(h; \dot{\mathcal{P}}_1) \leq 1 \cdot 2n\delta \leq \varepsilon/2$. For tags t_i in $\dot{\mathcal{P}}_2$, we have $h(t_i) < \varepsilon/2$ and the total length of the subintervals in $\dot{\mathcal{P}}_2$ is clearly less than 1, so that $S(h; \dot{\mathcal{P}}_2) < (\varepsilon/2) \cdot 1 = \varepsilon/2$. Therefore, combining these results, we get

$$0 \leq S(h; \dot{\mathcal{P}}) = S(h; \dot{\mathcal{P}}_1) + S(h; \dot{\mathcal{P}}_2) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we infer that $h \in \mathcal{R}[0, 1]$ with integral 0. □

Exercises for Section 7.1

1. If $I := [0, 4]$, calculate the norms of the following partitions:
   (a) $\mathcal{P}_1 := (0, 1, 2, 4)$, (b) $\mathcal{P}_2 := (0, 2, 3, 4)$,
   (c) $\mathcal{P}_3 := (0, 1, 1.5, 2, 3.4, 4)$, (d) $\mathcal{P}_4 := (0, .5, 2.5, 3.5, 4)$.

2. If $f(x) := x^2$ for $x \in [0, 4]$, calculate the following Riemann sums, where $\dot{\mathcal{P}}_i$ has the same partition points as in Exercise 1, and the tags are selected as indicated.
   (a) $\dot{\mathcal{P}}_1$ with the tags at the left endpoints of the subintervals.
   (b) $\dot{\mathcal{P}}_1$ with the tags at the right endpoints of the subintervals.
   (c) $\dot{\mathcal{P}}_2$ with the tags at the left endpoints of the subintervals.
   (d) $\dot{\mathcal{P}}_2$ with the tags at the right endpoints of the subintervals.

3. Show that $f : [a, b] \to \mathbb{R}$ is Riemann integrable on $[a, b]$ if and only if there exists $L \in \mathbb{R}$ such that for every $\varepsilon > 0$ there exists $\delta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ is any tagged partition with norm $||\dot{\mathcal{P}}|| \leq \delta_\varepsilon$, then $\left|S(f; \dot{\mathcal{P}}) - L\right| \leq \varepsilon$.

4. Let $\dot{\mathcal{P}}$ be a tagged partition of $[0, 3]$.
   (a) Show that the union U_1 of all subintervals in $\dot{\mathcal{P}}$ with tags in $[0, 1]$ satisfies $\left[0, 1 - \|\dot{\mathcal{P}}\|\right] \subseteq U_1 \subseteq \left[0, 1 + \|\dot{\mathcal{P}}\|\right]$.
   (b) Show that the union U_2 of all subintervals in $\dot{\mathcal{P}}$ with tags in $[1,2]$ satisfies $\left[1 + \|\dot{\mathcal{P}}\|, 2 - \|\dot{\mathcal{P}}\|\right] \subseteq U_2 \subseteq \left[1 - \|\dot{\mathcal{P}}\|, 2 + \|\dot{\mathcal{P}}\|\right]$.
5. Let $\dot{\mathcal{P}} := \{(I_i, t_i)\}_{i=1}^n$ be a tagged partition of $[a, b]$ and let $c_1 < c_2$.
   (a) If u belongs to a subinterval I_i whose tag satisfies $c_1 \leq t_i \leq c_2$, show that $c_1 - \|\dot{\mathcal{P}}\| \leq u \leq c_2 + \|\dot{\mathcal{P}}\|$.
   (b) If $v \in [a, b]$ and satisfies $c_1 + \|\dot{\mathcal{P}}\| \leq v \leq c_2 - \|\dot{\mathcal{P}}\|$, then the tag t_i of any subinterval I_i that contains v satisfies $t_i \in [c_1, c_2]$.
6. (a) Let $f(x) := 2$ if $0 \leq x < 1$ and $f(x) := 1$ if $1 \leq x \leq 2$. Show that $f \in \mathcal{R}[0, 2]$ and evaluate its integral.
   (b) Let $h(x) := 2$ if $0 \leq x < 1$, $h(1) := 3$ and $h(x) := 1$ if $1 < x \leq 2$. Show that $h \in \mathcal{R}[0, 2]$ and evaluate its integral.
7. Use Mathematical Induction and Theorem 7.1.5 to show that if $f_1, \ldots, f_n$ are in $\mathcal{R}[a, b]$ and if $k_1, \ldots, k_n \in \mathbb{R}$, then the linear combination $f = \sum_{i=1}^n k_i f_i$ belongs to $\mathcal{R}[a, b]$ and $\int_a^b f = \sum_{i=1}^n k_i \int_a^b f_i$.
8. If $f \in \mathcal{R}[a, b]$ and $|f(x)| \leq M$ for all $x \in [a, b]$, show that $\left|\int_a^b f\right| \leq M(b - a)$.
9. If $f \in \mathcal{R}[a, b]$ and if $(\dot{\mathcal{P}}_n)$ is any sequence of tagged partitions of $[a, b]$ such that $\|\dot{\mathcal{P}}_n\| \to 0$, prove that $\int_a^b f = \lim_n S(f; \dot{\mathcal{P}}_n)$.
10. Let $g(x) := 0$ if $x \in [0, 1]$ is rational and $g(x) := 1/x$ if $x \in [0, 1]$ is irrational. Explain why $g \notin \mathcal{R}[0, 1]$. However, show that there exists a sequence $(\dot{\mathcal{P}}_n)$ of tagged partitions of $[a, b]$ such that $\|\dot{\mathcal{P}}_n\| \to 0$ and $\lim_n S(g; \dot{\mathcal{P}}_n)$ exists.
11. Suppose that f is bounded on $[a, b]$ and that there exists two sequences of tagged partitions of $[a, b]$ such that $\|\dot{\mathcal{P}}_n\| \to 0$ and $\|\dot{\mathcal{Q}}_n\| \to 0$, but such that $\lim_n S(f; \dot{\mathcal{P}}_n) \neq \lim_n S(f; \dot{\mathcal{Q}}_n)$. Show that f is not in $\mathcal{R}[a, b]$.
12. Consider the Dirichlet function, introduced in Example 5.1.6(g), defined by $f(x) := 1$ for $x \in [0, 1]$ rational and $f(x) := 0$ for $x \in [0, 1]$ irrational. Use the preceding exercise to show that f is *not* Riemann integrable on $[0, 1]$.
13. Suppose that $c \leq d$ are points in $[a, b]$. If $\varphi : [a, b] \to \mathbb{R}$ satisfies $\varphi(x) = \alpha > 0$ for $x \in [c, d]$ and $\varphi(x) = 0$ elsewhere in $[a, b]$, prove that $\varphi \in \mathcal{R}[a, b]$ and that $\int_a^b \varphi = \alpha(d - c)$. [*Hint*: Given $\varepsilon > 0$ let $\delta_\varepsilon := \varepsilon/4\alpha$ and show that if $\|\dot{\mathcal{P}}\| < \delta_\varepsilon$ then we have $\alpha(d - c - 2\delta_\varepsilon) \leq S(\varphi; \dot{\mathcal{P}}) \leq \alpha(d - c + 2\delta_\varepsilon)$.]
14. Let $0 \leq a < b$, let $Q(x) := x^2$ for $x \in [a, b]$ and let $\mathcal{P} := \{[x_{i-1}, x_i]\}_{i=1}^n$ be a partition of $[a, b]$. For each i, let q_i be the positive *square root* of
$$\tfrac{1}{3}\left(x_i^2 + x_i x_{i-1} + x_{i-1}^2\right).$$
   (a) Show that q_i satisfies $0 \leq x_{i-1} \leq q_i \leq x_i$.
   (b) Show that $Q(q_i)(x_i - x_{i-1}) = \frac{1}{3}\left(x_i^3 - x_{i-1}^3\right)$.
   (c) If $\dot{\mathcal{Q}}$ is the tagged partition with the same subintervals as $\mathcal{P}$ and the tags q_i, show that $S(Q; \dot{\mathcal{Q}}) = \frac{1}{3}\left(b^3 - a^3\right)$.
   (d) Use the argument in Example 7.1.4(c) to show that $Q \in \mathcal{R}[a, b]$ and
$$\int_a^b Q = \int_a^b x^2 dx = \tfrac{1}{3}\left(b^3 - a^3\right).$$
15. If $f \in \mathcal{R}[a, b]$ and $c \in \mathbb{R}$, we define g on $[a + c, b + c]$ by $g(y) := f(y - c)$. Prove that $g \in \mathcal{R}[a + c, b + c]$ and that $\int_{a+c}^{b+c} g = \int_a^b f$. The function g is called the c-**translate** of f.

Section 7.2 Riemann Integrable Functions

We begin with a proof of the important Cauchy Criterion. We will then prove the Squeeze Theorem, which will be used to establish the Riemann integrability of several classes of functions (step functions, continuous functions, and monotone functions). Finally we will establish the Additivity Theorem.

We have already noted that direct use of the definition requires that we know the value of the integral. The Cauchy Criterion removes this need, but at the cost of considering two Riemann sums, instead of just one.

7.2.1 Cauchy Criterion *A function:* $[a, b] \to \mathbb{R}$ *belongs to* $\mathcal{R}[a,b]$ *if and only if for every* $\varepsilon > 0$ *there exists* $\eta_\varepsilon > 0$ *such that if* $\dot{\mathcal{P}}$ *and* $\dot{\mathcal{Q}}$ *are any tagged partitions of* $[a, b]$ *with* $||\dot{\mathcal{P}}|| < \eta_\varepsilon$ *and* $||\dot{\mathcal{Q}}|| < \eta_\varepsilon$, *then*

$$\left|S(f;\dot{\mathcal{P}}) - S(f;\dot{\mathcal{Q}})\right| < \varepsilon. \qquad \square$$

Proof. ($\Rightarrow$) If $f \in \mathcal{R}[a, b]$ with integral L, let $\eta_\varepsilon := \delta_{\varepsilon/2} > 0$ be such that if $\dot{\mathcal{P}}$, $\dot{\mathcal{Q}}$ are tagged partitions such that $||\dot{\mathcal{P}}|| < \eta_\varepsilon$ and $||\dot{\mathcal{Q}}|| < \eta_\varepsilon$, then

$$\left|S(f;\dot{\mathcal{P}}) - L\right| < \varepsilon/2 \quad \text{and} \quad \left|S(f;\dot{\mathcal{Q}}) - L\right| < \varepsilon/2.$$

Therefore we have

$$\begin{aligned}\left|S(f;\dot{\mathcal{P}}) - S(f;\dot{\mathcal{Q}})\right| &\le \left|S(f;\dot{\mathcal{P}}) - L + L - S(f;\dot{\mathcal{Q}})\right| \\ &\le \left|S(f;\dot{\mathcal{P}}) - L\right| + \left|L - S(f;\dot{\mathcal{Q}})\right| \\ &< \varepsilon/2 + \varepsilon/2 = \varepsilon.\end{aligned}$$

($\Leftarrow$) For each $n \in \mathbb{N}$, let $\delta_n > 0$ be such that if $\dot{\mathcal{P}}$ and $\dot{\mathcal{Q}}$ are tagged partitions with norms $< \delta_n$, then

$$\left|S(f;\dot{\mathcal{P}}) - f(f;\dot{\mathcal{Q}})\right| < 1/n.$$

Evidently we may assume that $\delta_n \ge \delta_{n+1}$ for $n \in \mathbb{N}$; otherwise, we replace δ_n by $\delta_n' := \min\{\delta_1, \ldots, \delta_n\}$.

For each $n \in \mathbb{N}$, let $\dot{\mathcal{P}}_n$ be a tagged partition with $||\dot{\mathcal{P}}_n|| < \delta_n$. Clearly, if $m > n$ then both $\dot{\mathcal{P}}_m$ and $\dot{\mathcal{P}}_n$ have norms $< \delta_n$, so that

$$\left|S(f;\dot{\mathcal{P}}_n) - S(f;\dot{\mathcal{P}}_m)\right| < 1/n \qquad \text{for} \qquad m > n. \tag{1}$$

Consequently, the sequence $\left(S(f;\dot{\mathcal{P}}_m)\right)_{m=1}^{\infty}$ is a Cauchy sequence in $\mathbb{R}$. Therefore (by Theorem 3.5.5) this sequence converges in $\mathbb{R}$ and we let $A := \lim_m S(f;\dot{\mathcal{P}}_m)$.

Passing to the limit in (1) as $m \to \infty$, we have

$$\left|S(f;\dot{\mathcal{P}}_n) - A\right| \le 1/n \qquad \text{for all} \qquad n \in \mathbb{N}.$$

To see that A is the Riemann integral of f, given $\varepsilon > 0$, let $K \in \mathbb{N}$ satisfy $K > 2/\varepsilon$. If $\dot{\mathcal{Q}}$ is any tagged partition with $||\dot{\mathcal{Q}}|| < \delta_K$, then

$$\begin{aligned}\left|S(f;\dot{\mathcal{Q}}) - A\right| &\le \left|S(f;\dot{\mathcal{Q}}) - S(f;\dot{\mathcal{P}}_K)\right| + \left|S(f;\dot{\mathcal{P}}_K) - A\right| \\ &\le 1/K + 1/K < \varepsilon.\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, then $f \in \mathcal{R}[a, b]$ with integral A. Q.E.D.

We will now give two examples of the use of the Cauchy Criterion.

7.2.2 Examples **(a)** Let $g : [0,3] \to \mathbb{R}$ be the function considered in Example 7.1.4(b). In that example we saw that if $\dot{\mathcal{P}}$ is a tagged partition of $[0, 3]$ with norm $||\dot{\mathcal{P}}|| < \delta$, then

$$8 - 5\delta \leq S(g;\dot{\mathcal{P}}) \leq 8 + 5\delta.$$

Hence if $\dot{\mathcal{Q}}$ is another tagged partition with $||\dot{\mathcal{Q}}|| < \delta$, then

$$8 - 5\delta \leq S(g;\dot{\mathcal{Q}}) \leq 8 + 5\delta.$$

If we subtract these two inequalities, we obtain

$$\left|S(g;\dot{\mathcal{P}}) - S(g;\dot{\mathcal{Q}})\right| \leq 10\delta.$$

In order to make this final term $< \varepsilon$, we are led to employ the Cauchy Criterion with $\eta_\varepsilon := \varepsilon/20$. (We leave the details to the reader.)

(b) The Cauchy Criterion can be used to show that a function $f : [a,b] \to \mathbb{R}$ is *not* Riemann integrable. To do this we need to show that: *There exists* $\varepsilon_0 > 0$ *such that for any* $\eta > 0$ *there exists tagged partitions* $\dot{\mathcal{P}}$ *and* $\dot{\mathcal{Q}}$ *with* $||\dot{\mathcal{P}}|| < \eta$ *and* $||\dot{\mathcal{Q}}|| < \eta$ *such that* $\left|S(f;\dot{\mathcal{P}}) - S(f;\dot{\mathcal{Q}})\right| \geq \varepsilon_0$.

We will apply these remarks to the Dirichlet function, considered in 5.1.6(g), defined by $f(x) := 1$ if $x \in [0,1]$ is rational and $f(x) := 0$ if $x \in [0,1]$ is irrational.

Here we take $\varepsilon_0 := \frac{1}{2}$. If $\dot{\mathcal{P}}$ is any partition all of whose tags are rational numbers then $S(f;\dot{\mathcal{P}}) = 1$, while if $\dot{\mathcal{Q}}$ is any tagged partition all of whose tags are irrational numbers then $S(f;\dot{\mathcal{Q}}) = 0$. Since we are able to take such tagged partitions with arbitrarily small norms, we conclude that the Dirichlet function is *not* Riemann integrable. □

The Squeeze Theorem

In working with the definition of Riemann integral, we have encountered two types of difficulties. First, for each partition, there are infinitely many choices of tags. And second, there are infinitely many partitions that have a norm less than a specified amount. We have experienced dealing with these difficulties in examples and proofs of theorems. We will now establish an important tool for proving integrability called the Squeeze Theorem that will provide some relief from those difficulties. It states that if a given function can be "squeezed" or bracketed between two functions that are known to be Riemann integrable with sufficient accuracy, then we may conclude that the given function is also Riemann integrable. The exact conditions are given in the statement of the theorem. (The idea of squeezing a function to establish integrability led the French mathematician Gaston Darboux to develop an approach to integration by means of upper and lower integrals, and this approach is presented in Section 7.4.)

7.2.3 Squeeze Theorem *Let* $f : [a,b] \to \mathbb{R}$. *Then* $f \in \mathcal{R}[a,b]$ *if and only if for every* $\varepsilon > 0$ *there exist functions* α_ε *and* ω_ε *in* $\mathcal{R}[a,b]$ *with*

$$\alpha_\varepsilon(x) \leq f(x) \leq \omega_\varepsilon(x) \qquad \text{for all} \qquad x \in [a,b], \tag{2}$$

and such that

$$\int_a^b (\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon. \tag{3}$$

Proof. ($\Rightarrow$) Take $\alpha_\varepsilon = \omega_\varepsilon = f$ for all $\varepsilon > 0$.

($\Leftarrow$) Let $\varepsilon > 0$. Since α_ε and ω_ε belong to $\mathcal{R}[a,b]$, there exists $\delta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ is any tagged partition with $||\dot{\mathcal{P}}|| < \delta_\varepsilon$ then

$$\left|S(\alpha_\varepsilon;\dot{\mathcal{P}}) - \int_a^b \alpha_\varepsilon\right| < \varepsilon \qquad \text{and} \qquad \left|S(\omega_\varepsilon;\dot{\mathcal{P}}) - \int_a^b \omega_\varepsilon\right| < \varepsilon.$$

It follows from these inequalities that

$$\int_a^b \alpha_\varepsilon - \varepsilon < S(\alpha_\varepsilon;\dot{\mathcal{P}}) \qquad \text{and} \qquad S(\omega_\varepsilon;\dot{\mathcal{P}}) < \int_a^b \omega_\varepsilon + \varepsilon.$$

In view of inequality (2), we have $S(\alpha_\varepsilon;\dot{\mathcal{P}}) \leq S(f;\dot{\mathcal{P}}) \leq S(\omega_\varepsilon;\dot{\mathcal{P}})$, whence

$$\int_a^b \alpha_\varepsilon - \varepsilon < S(f;\dot{\mathcal{P}}) < \int_a^b \omega_\varepsilon + \varepsilon.$$

If $\dot{\mathcal{Q}}$ is another tagged partition with $||\dot{\mathcal{Q}}|| < \delta_\varepsilon$, then we also have

$$\int_a^b \alpha_\varepsilon - \varepsilon < S(f;\dot{\mathcal{Q}}) < \int_a^b \omega_\varepsilon + \varepsilon.$$

If we subtract these two inequalities and use (3), we conclude that

$$\begin{aligned} |S(f;\dot{\mathcal{P}}) - S(f;\dot{\mathcal{Q}})| &< \int_a^b \omega_\varepsilon - \int_a^b \alpha_\varepsilon + 2\varepsilon \\ &= \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) + 2\varepsilon < 3\varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, the Cauchy Criterion implies that $f \in \mathcal{R}[a,b]$. Q.E.D.

Classes of Riemann Integrable Functions

The Squeeze Theorem is often used in connection with the class of step functions. It will be recalled from Definition 5.4.9 that a function $\varphi : [a,b] \to \mathbb{R}$ is a **step function** if it has only a finite number of distinct values, each value being assumed on one or more subintervals of $[a,b]$. For illustrations of step functions, see Figures 5.4.3 or 7.1.4.

7.2.4 Lemma If J *is a subinterval of* $[a,b]$ *having endpoints* $c < d$ *and if* $\varphi_J(x) := 1$ *for* $x \in J$ *and* $\varphi_J(x) := 0$ *elsewhere in* $[a,b]$, *then* $\varphi_J \in \mathcal{R}[a,b]$ *and* $\int_a^b \varphi_J = d - c$.

Proof. If $J = [c,d]$ with $c \leq d$, this is Exercise 7.1.13 and we can choose $\delta_\varepsilon := \varepsilon/4$.

There are three other subintervals J having the same endpoints c and d, namely, $[c,d)$, $(c,d]$, and (c,d). Since, by Theorem 7.1.3, we can change the value of a function at finitely many points without changing the integral, we have the same result for these other three subintervals. Therefore, we conclude that all four functions φ_J are integrable with integral equal to $d - c$. Q.E.D.

It is an important fact that any step function is Riemann integrable.

7.2.5 Theorem *If* $\varphi : [a,b] \to \mathbb{R}$ *is a step function, then* $\varphi \in \mathcal{R}[a,b]$.

Proof. Step functions of the type appearing in 7.2.4 are called "elementary step functions." In Exercise 5 it is shown that an arbitrary step function φ can be expressed

as a linear combination of such elementary step functions:

$$\varphi = \sum_{j=1}^{m} k_j \varphi_{J_j}, \tag{4}$$

where J_j has endpoints $c_j < d_j$. The lemma and Theorem 7.1.5(a, b) imply that $\varphi \in \mathcal{R}[a, b]$ and that

$$\int_a^b \varphi = \sum_{j=1}^{m} k_j (d_j - c_j). \tag{5}$$

Q.E.D.

We illustrate the use of step functions and the Squeeze Theorem in the next two examples. The first reconsiders a function that originally required a complicated calculation.

7.2.6 Examples **(a)** The function g in Example 7.1.4(b) is defined by $g(x) = 2$ for $0 \le x \le 1$ and $g(x) = 3$ for $1 < x \le 3$. We now see that g is a step function and therefore we calculate its integral to be $\int_0^3 g = 2 \cdot (1 - 0) + 3 \cdot (3 - 1) = 2 + 6 = 8$.

(b) Let $h(x) := x$ on [0,1] and let $P_n := (0, 1/n, 2/n, \ldots, (n-1)/n, n/n = 1)$. We define the step functions α_n and ω_n on the disjoint subintervals $[0, 1/n)$, $[1/n, 2/n)$, $\ldots, [(n-2)/n), (n-1)/n)$, $[(n-1)/n, 1]$ as follows: $\alpha_n(x) := h((k-1)/n) = (k-1)/n$ for x in $[(k-1)/n, k/n)$ for $k = 1, 2, \ldots, n-1$, and $\alpha_n(x) := h((n-1)/n) = (n-1)/n$ for x in $[(n-1)/n, 1]$. That is, α_n has the minimum value of h on each subinterval. Similarly, we define ω_n to be the maximum value of h on each subinterval, that is, $\omega_n(x) := k/n$ for x in $[(k-1)/n, k/n)$ for $k = 1, 2, \ldots, n-1$, and $\omega_n(x) := 1$ for x in $[(n-1)/n, 1]$. (The reader should draw a sketch for the case $n = 4$.)

Then we get

$$\begin{aligned} \int_0^1 \alpha_n &= \frac{1}{n}(0 + 1/n + 2/n + \cdots + (n-1)/n) \\ &= \frac{1}{n^2}(1 + 2 + \cdots + (n-1)) \\ &= \frac{1}{n^2} \cdot \frac{(n-1)n}{2} = \frac{1}{2}(1 - 1/n). \end{aligned}$$

In a similar manner, we also get $\int_0^1 \omega_n = \frac{1}{2}(1 + 1/n)$. Thus we have

$$\alpha_n(x) \le h(x) \le \omega_n(x)$$

for $x \in [0, 1]$ and

$$\int_0^1 (\omega_n - \alpha_n) = \frac{1}{n}.$$

Since for a given $\varepsilon > 0$, we can choose n so that $\frac{1}{n} < \varepsilon$, it follows from the Squeeze Theorem that h is integrable. We also see that the value of the integral of h lies between the integrals of α_n and ω_n for all n and therefore has value $\frac{1}{2}$. □

We will now use the Squeeze Theorem to show that an arbitrary continuous function is Riemann integrable.

7.2.7 Theorem *If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, then $f \in \mathcal{R}[a, b]$.*

Proof. It follows from Theorem 5.4.3 that f is uniformly continuous on $[a, b]$. Therefore, given $\varepsilon > 0$ there exists $\delta_\varepsilon > 0$ such that if $u, v \in [a, b]$ and $|u - v| < \delta_\varepsilon$, then we have $|f(u) - f(v)| < \varepsilon/(b - a)$.

Let $\mathcal{P} = \{I_i\}_{i=1}^n$ be a partition such that $||\mathcal{P}|| < \delta_\varepsilon$. Applying Theorem 5.3.4 we let $u_i \in I_i$ be a point where f attains its minimum value on I_i, and let $v_i \in I_i$ be a point where f attains its maximum value on I_i.

Let α_ε be the step function defined by $\alpha_\varepsilon(x) := f(u_i)$ for $x \in [x_{i-1}, x_i)$ $(i = 1, \ldots, n - 1)$ and $\alpha_\varepsilon(x) := f(u_n)$ for $x \in [x_{n-1}, x_n]$. Let ω_ε be defined similarly using the points v_i instead of the u_i. Then one has

$$\alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x) \qquad \text{for all} \qquad x \in [a, b].$$

Moreover, it is clear that

$$\begin{aligned} 0 \le \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) &= \sum_{i=1}^n (f(v_i) - f(u_i))(x_i - x_{i-1}) \\ &< \sum_{i=1}^n \left(\frac{\varepsilon}{b - a}\right)(x_i - x_{i-1}) = \varepsilon. \end{aligned}$$

Therefore it follows from the Squeeze Theorem that $f \in \mathcal{R}[a, b]$. Q.E.D.

Monotone functions are not necessarily continuous at every point, but they are also Riemann integrable.

7.2.8 Theorem *If $f : [a, b] \to \mathbb{R}$ is monotone on $[a, b]$, then $f \in \mathcal{R}[a, b]$.*

Proof. Assume that f is increasing on $I = [a, b]$. Partitioning the interval into n equal subintervals $I_k = [x_{k-1}, x_k]$ gives us $x_k - x_{k-1} = (b - a)/n$, $k = 1, 2, \ldots, n$. Since f is increasing on I_k, its minimum value is attained at the left endpoint x_{k-1} and its maximum value is attained at the right endpoint x_k. Therefore, we define the step functions $\alpha(x) := f(x_{k-1})$ and $\omega(x) := f(x_k)$ for $x \in [x_{k-1}, x_k)$, $k = 1, 2, \ldots, n - 1$, and $\alpha(x) := f(x_{n-1})$ and $\omega(x) := f(x_n)$ for $x \in [x_{n-1}, x_n]$. Then we have $\alpha(x) \le f(x) \le \omega(x)$ for all $x \in I$, and

$$\int_a^b \alpha = \frac{b - a}{n}(f(x_0) + f(x_1) + \cdots + f(x_{n-1}))$$

$$\int_a^b \omega = \frac{b - a}{n}(f(x_1) + \cdots + f(x_{n-1}) + f(x_n)).$$

Subtracting, and noting the many cancellations, we obtain

$$\int_a^b (\omega - \alpha) = \frac{b - a}{n}(f(x_n) - f(x_0)) = \frac{b - a}{n}(f(b) - f(a)).$$

Thus for a given $\varepsilon > 0$, we choose n such that $n > (b - a)(f(b) - f(a))/\varepsilon$. Then we have $\int_a^b (\omega - \alpha) < \varepsilon$ and the Squeeze Theorem implies that f is integrable on I. Q.E.D.

The Additivity Theorem

We now return to arbitrary Riemann integrable functions. Our next result shows that the integral is an "additive function" of the interval over which the function is integrated. This property is no surprise, but its proof is a bit delicate and may be omitted on a first reading.

7.2.9 Additivity Theorem *Let $f := [a,b] \to \mathbb{R}$ and let $c \in (a,b)$. Then $f \in \mathcal{R}[a,b]$ if and only if its restrictions to $[a, c]$ and $[c, b]$ are both Riemann integrable. In this case*

$$\int_a^b f = \int_a^c f + \int_c^b f. \tag{6}$$

Proof. ($\Leftarrow$) Suppose that the restriction f_1 of f to $[a, c]$, and the restriction f_2 of f to $[c, b]$ are Riemann integrable to L_1 and L_2 respectively. Then, given $\varepsilon > 0$ there exists $\delta' > 0$ such that if $\dot{\mathcal{P}}_1$ is a tagged partition of $[a, c]$ with $||\dot{\mathcal{P}}_1|| < \delta'$, then $\left|S(f_1; \dot{\mathcal{P}}_1) - L_1\right| < \varepsilon/3$. Also there exists $\delta'' > 0$ such that if $\dot{\mathcal{P}}_2$ is a tagged partition of $[c, b]$ wilh $||\dot{\mathcal{P}}_2|| < \delta''$ then $\left|S(f_2; \dot{\mathcal{P}}_2) - L_2\right| < \varepsilon/3$. If M is a bound for $|f|$, we define $\delta_\varepsilon := \min\{\delta', \delta'', \varepsilon/6M\}$ and let $\dot{\mathcal{P}}$ be a tagged partition of $[a, b]$ with $||\dot{\mathcal{Q}}|| < \delta$. We will prove that

$$\left|S(f;\dot{\mathcal{Q}}) - (L_1 + L_2)\right| < \varepsilon. \tag{7}$$

(i) If c is a partition point of $\dot{\mathcal{Q}}$, we split $\dot{\mathcal{Q}}$ into a partition $\dot{\mathcal{Q}}_1$ of $[a, c]$ and a partition $\dot{\mathcal{Q}}_2$ of $[c, b]$. Since $S(f;\dot{\mathcal{Q}}) = S(f;\dot{\mathcal{Q}}_1) + S(f;\dot{\mathcal{Q}}_2)$, and since $\dot{\mathcal{Q}}_1$ has norm $< \delta'$ and $\dot{\mathcal{Q}}_2$ has norm $< \delta''$, the inequality (7) is clear.

(ii) If c is not a partition point in $\dot{\mathcal{Q}} = \{(I_k, t_k)\}_{k=1}^m$, there exists $k \le m$ such that $c \in (x_{k-1}, x_k)$. We let $\dot{\mathcal{Q}}_1$ be the tagged partition of $[a, c]$ defined by

$$\dot{\mathcal{Q}}_1 := \{(I_1, t_1), \ldots, (I_{k-1}, t_{k-1}), ([x_{k-1}, c], c)\},$$

and $\dot{\mathcal{Q}}_2$ be the tagged partition of $[c, b]$ defined by

$$\dot{\mathcal{Q}}_2 := \{([c, x_k], c), (I_{k+1}, t_{k+1}), \ldots, (I_m, t_m)\}.$$

A straightforward calculation shows that

$$\begin{aligned} S(f;\dot{\mathcal{Q}}) - S(f;\dot{\mathcal{Q}}_1) - S(f;\dot{\mathcal{Q}}_2) &= f(t_k)(x_k - x_{k-1}) - f(c)(x_k - x_{k-1}) \\ &= (f(t_k) - f(c)) \cdot (x_k - x_{k-1}), \end{aligned}$$

whence it follows that

$$\left|S(f;\dot{\mathcal{Q}}) - S(f;\dot{\mathcal{Q}}_1) - S(f;\dot{\mathcal{Q}}_2)\right| \le 2M(x_k - x_{k-1}) < \varepsilon/3.$$

But since $||\dot{\mathcal{Q}}_1|| < \delta \le \delta'$ and $||\dot{\mathcal{Q}}_2|| < \delta \le \delta''$, it follows that

$$\left|S(f;\dot{\mathcal{Q}}_1) - L_1\right| < \varepsilon/3 \qquad \text{and} \qquad \left|S(f;\dot{\mathcal{Q}}_2) - L_2\right| < \varepsilon/3,$$

from which we obtain (7). Since $\varepsilon > 0$ is arbitrary, we infer that $f \in \mathcal{R}[a,b]$ and that (6) holds.

($\Rightarrow$) We suppose that $f \in \mathcal{R}[a,b]$ and, given $\varepsilon > 0$, we let $\eta_\varepsilon > 0$ satisfy the Cauchy Criterion 7.2.1. Let f_1 be the restriction of f to $[a, c]$ and let $\dot{\mathcal{P}}_1$, $\dot{\mathcal{Q}}_1$ be tagged partitions of

$[a, c]$ with $||\dot{\mathcal{P}}_1|| < \eta_\varepsilon$ and $||\dot{\mathcal{Q}}_1|| < \eta_\varepsilon$. By adding additional partition points and tags from $[c, b]$, we can extend $\dot{\mathcal{P}}_1$ and $\dot{\mathcal{Q}}_1$ to tagged partitions $\dot{\mathcal{P}}$ and $\dot{\mathcal{Q}}$ of $[a, b]$ that satisfy $||\dot{\mathcal{P}}|| < \eta_\varepsilon$ and $||\dot{\mathcal{Q}}|| < \eta_\varepsilon$. If we use the *same* additional points and tags in $[c, b]$ for both $\dot{\mathcal{P}}$ and $\dot{\mathcal{Q}}$, then

$$S(f_1; \dot{\mathcal{P}}_1) - S(f_1; \dot{\mathcal{Q}}_1) = S(f; \dot{\mathcal{P}}) - S(f; \dot{\mathcal{Q}}).$$

Since both $\dot{\mathcal{P}}$ and $\dot{\mathcal{Q}}$ have norm η_ε, then $|S(f_1; \dot{\mathcal{P}}_1) - S(f_1; \dot{\mathcal{Q}}_1)| < \varepsilon$. Therefore the Cauchy Condition shows that the restriction f_1 of f to $[a, c]$ is in $\mathcal{R}[a, c]$. In the same way, we see that the restriction f_2 of f to $[c, b]$ is in $\mathcal{R}[c, d]$.

The equality (6) now follows from the first part of the theorem. Q.E.D.

7.2.10 Corollary *If $f \in \mathcal{R}[a, b]$, and if $[c, d] \subseteq [a, b]$, then the restriction of f to $[c, d]$ is in $\mathcal{R}[c, d]$.*

Proof. Since $f \in \mathcal{R}[a, b]$ and $c \in [a, b]$, it follows from the theorem that its restriction to $[c, b]$ is in $\mathcal{R}[c, b]$. But if $d \in [c, b]$, then another application of the theorem shows that the restriction of f to $[c, d]$ is in $\mathcal{R}[c, d]$. Q.E.D.

7.2.11 Corollary *If $f \in \mathcal{R}[a, b]$ and if $a = c_0 < c_1 < \cdots < c_m = b$, then the restrictions of f to each of the subintervals $[c_{i-1}, c_i]$ are Riemann integrable and*

$$\int_a^b f = \sum_{i=1}^{m} \int_{c_{i-1}}^{c_i} f.$$

Until now, we have considered the Riemann integral over an interval $[a, b]$ where $a < b$. It is convenient to have the integral defined more generally.

7.2.12 Definition If $f \in \mathcal{R}[a, b]$ and if $\alpha, \beta \in [a, b]$ with $\alpha < \beta$, we define

$$\int_\beta^\alpha f := -\int_\alpha^\beta f \qquad \text{and} \qquad \int_\alpha^\alpha f := 0.$$

7.2.13 Theorem *If $f \in \mathcal{R}[a, b]$ and if α, β, γ are any numbers in $[a, b]$, then*

$$\int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f, \tag{8}$$

in the sense that the existence of any two of these integrals implies the existence of the third integral and the equality (8).

Proof. If any two of the numbers α, β, γ are equal, then (8) holds. Thus we may suppose that all three of these numbers are distinct.

For the sake of symmetry, we introduce the expression

$$L(\alpha, \beta, \gamma) := \int_\alpha^\beta f + \int_\beta^\gamma f + \int_\gamma^\alpha f.$$

It is clear that (8) holds if and only if $L(\alpha, \beta, \gamma) = 0$. Therefore, to establish the assertion, we need to show that $L = 0$ for all six permutations of the arguments α, β, and γ.

We note that the Additivity Theorem 7.2.9 implies that $L(\alpha, \beta, \gamma) = 0$ when $\alpha < \gamma < \beta$. But it is easily seen that both $L(\beta, \gamma, \alpha)$ and $L(\gamma, \alpha, \beta)$ equal $L(\alpha, \beta, \gamma)$.

Moreover, the numbers

$$L(\beta, \alpha, \gamma), \qquad L(\alpha, \gamma, \beta), \qquad \text{and} \qquad L(\gamma, \beta, \alpha)$$

are all equal to $-L(\alpha, \beta, \gamma)$. Therefore, L vanishes for all possible configurations of these three points. Q.E.D.

Exercises for Section 7.2

1. Let $f : [a, b] \to \mathbb{R}$. Show that $f \notin \mathcal{R}[a, b]$ if and only if there exists $\varepsilon_0 > 0$ such that for every $n \in \mathbb{N}$ there exist tagged partitions $\dot{\mathcal{P}}_n$ and $\dot{\mathcal{Q}}_n$ with $||\dot{\mathcal{P}}_n|| < 1/n$ and $||\dot{\mathcal{Q}}_n|| < 1/n$ such that $\left|S(f; \dot{\mathcal{P}}_n) - S(f; \dot{\mathcal{Q}}_n)\right| \geq \varepsilon_0$.
2. Consider the function h defined by $h(x) := x + 1$ for $x \in [0, 1]$ rational, and $h(x) := 0$ for $x \in [0, 1]$ irrational. Show that h is not Riemann integrable.
3. Let $H(x) := k$ for $x = 1/k$ ($k \in \mathbb{N}$) and $H(x) := 0$ elsewhere on [0, 1]. Use Exercise 1, or the argument in 7.2.2(b), to show that H is not Riemann integrable.
4. If $\alpha(x) := -x$ and $\omega(x) := x$ and if $\alpha(x) \leq f(x) \leq \omega(x)$ for all $x \in [0, 1]$, does it follow from the Squeeze Theorem 7.2.3 that $f \in \mathcal{R}[0, 1]$?
5. If J is any subinterval of $[a, b]$ and if $\varphi_J(x) := 1$ for $x \in J$ and $\varphi_J(x) := 0$ elsewhere on $[a, b]$, we say that φ_J is an *elementary step function* on $[a, b]$. Show that every step function is a linear combination of elementary step functions.
6. If $\psi : [a, b] \to \mathbb{R}$ takes on only a finite number of distinct values, is ψ a step function?
7. If $S(f; \dot{\mathcal{P}})$ is any Riemann sum of $f : [a, b] \to \mathbb{R}$, show that there exists a step function $\varphi : [a, b] \to \mathbb{R}$ such that $\int_a^b \varphi = S(f; \dot{\mathcal{P}})$.
8. Suppose that f is continuous on $[a, b]$, that $f(x) \geq 0$ for all $x \in [a, b]$ and that $\int_a^b f = 0$. Prove that $f(x) = 0$ for all $x \in [a, b]$.
9. Show that the continuity hypothesis in the preceding exercise cannot be dropped.
10. If f and g are continuous on $[a, b]$ and if $\int_a^b f = \int_a^b g$, prove that there exists $c \in [a, b]$ such that $f(c) = g(c)$.
11. If f is bounded by M on $[a, b]$ and if the restriction of f to every interval $[c, b]$ where $c \in (a, b)$ is Riemann integrable, show that $f \in \mathcal{R}[a, b]$ and that $\int_c^b f \to \int_a^b f$ as $c \to a+$. [*Hint:* Let $\alpha_c(x) := -M$ and $\omega_c(x) := M$ for $x \in [a, c)$ and $\alpha_c(x) := \omega_c(x) := f(x)$ for $x \in [c, b]$. Apply the Squeeze Theorem 7.2.3 for c sufficiently near a.]
12. Show that $g(x) := \sin(1/x)$ for $x \in (0, 1]$ and $g(0) := 0$ belongs to $\mathcal{R}[0, 1]$.
13. Give an example of a function $f : [a, b] \to \mathbb{R}$ that is in $\mathcal{R}[c, b]$ for every $c \in (a, b)$ but which is not in $\mathcal{R}[a, b]$.
14. Suppose that $f : [a, b] \to \mathbb{R}$, that $a = c_0 < c_1 < \cdots < c_m = b$ and that the restrictions of f to $[c_{i-1}, c_i]$ belong to $\mathcal{R}[c_{i-1}, c_i]$ for $i = 1, \ldots, m$. Prove that $f \in \mathcal{R}[a, b]$ and that the formula in Corollary 7.2.11 holds.
15. If f is bounded and there is a finite set E such that f is continuous at every point of $[a, b] \backslash E$, show that $f \in \mathcal{R}[a, b]$.
16. If f is continuous on $[a, b]$, $a < b$, show that there exists $c \in [a, b]$ such that we have $\int_a^b f = f(c)(b - a)$. This result is sometimes called the *Mean Value Theorem for Integrals*.
17. If f and g are continuous on $[a, b]$ and $g(x) > 0$ for all $x \in [a, b]$, show that there exists $c \in [a, b]$ such that $\int_a^b fg = f(c) \int_a^b g$. Show that this conclusion fails if we do not have $g(x) > 0$. (Note that this result is an extension of the preceding exercise.)

18. Let f be continuous on $[a, b]$, let $f(x) \geq 0$ for $x \in [a, b]$, and let $M_n := (\int_a^b f^n)^{1/n}$. Show that $\lim(M_n) = \sup\{f(x) : x \in [a, b]\}$.
19. Suppose that $a > 0$ and that $f \in \mathcal{R}[-a, a]$.
    (a) If f is *even* (that is, if $f(-x) = f(x)$ for all $x \in [0, a]$), show that $\int_{-a}^{a} f = 2\int_0^a f$.
    (b) If f is *odd* (that is, if $f(-x) = -f(x)$ for all $x \in [0, a]$), show that $\int_{-a}^{a} f = 0$.
20. If f is continuous on $[-a, a]$, show that $\int_{-a}^{a} f(x^2)dx = 2\int_0^a f(x^2)dx$.

Section 7.3 The Fundamental Theorem

We will now explore the connection between the notions of the derivative and the integral. In fact, there are *two* theorems relating to this problem: one has to do with integrating a derivative, and the other with differentiating an integral. These theorems, taken together, are called the Fundamental Theorem of Calculus. Roughly stated, they imply that the operations of differentiation and integration are inverse to each other. However, there are some subtleties that should not be overlooked.

The Fundamental Theorem (First Form)

The First Form of the Fundamental Theorem provides a theoretical basis for the method of calculating an integral that the reader learned in calculus. It asserts that if a function f is the derivative of a function F, *and if* f *belongs to* $\mathcal{R}[a, b]$, then the integral $\int_a^b f$ can be calculated by means of the evaluation $F|_a^b := F(b) - F(a)$. A function F such that $F'(x) = f(x)$ for all $x \in [a, b]$ is called an **antiderivative** or a **primitive of** f on $[a, b]$. Thus, when f has an antiderivative, it is a very simple matter to calculate its integral.

In practice, it is convenient to allow some exceptional points c where $F'(c)$ does not exist in $\mathbb{R}$, or where it does not equal $f(c)$. It turns out that we can permit a *finite* number of such exceptional points.

7.3.1 Fundamental Theorem of Calculus (First Form) *Suppose there is a* ***finite*** *set* $\boldsymbol{E}$ *in* $[a, b]$ *and functions* f, $F := [a, b] \to \mathbb{R}$ *such that*:

(a) F *is continuous on* $[a, b]$,

(b) $F'(x) = f(x)$ *for all* $x \in [a, b] \backslash E$,

(c) f *belongs to* $\mathcal{R}[a, b]$.

Then we have

$$\int_a^b f = F(b) - F(a). \tag{1}$$

Proof. We will prove the theorem in the case where $E := \{a, b\}$. The general case can be obtained by breaking the interval into the union of a finite number of intervals (see Exercise 1).

Let $\varepsilon > 0$ be given. Since $f \in \mathcal{R}[a, b]$ by assumption (c), there exists $\delta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ is *any* tagged partition with $||\dot{\mathcal{P}}|| < \delta_\varepsilon$, then

$$\left| S(f; \dot{\mathcal{P}}) - \int_a^b f \right| < \varepsilon. \tag{2}$$

If the subintervals in $\dot{\mathcal{P}}$ are $[x_{i-1}, x_i]$, then the Mean Value Theorem 6.2.4 applied to F on $[x_{i-1}, x_i]$ implies that there exists $u_i \in (x_{i-1}, x_i)$ such that

$$F(x_i) - F(x_{i-1}) = F'(u_i) \cdot (x_i - x_{i-1}) \qquad \text{for} \qquad i = 1, \ldots, n.$$

If we add these terms, note the telescoping of the sum, and use the fact that $F'(u_i) = f(u_i)$, we obtain

$$F(b) - F(a) = \sum_{i=1}^{n} (F(x_i) - F(x_{i-1})) = \sum_{i=1}^{n} f(u_i)(x_i - x_{i-1}).$$

Now let $\dot{\mathcal{P}}_u := \{([x_{i-1}, x_i], u_i)\}_{i=1}^{n}$, so the sum on the right equals $S(f; \dot{\mathcal{P}}_u)$. If we substitute $F(b) - F(a) = S(f; \dot{\mathcal{P}}_u)$ into (2), we conclude that

$$\left| F(b) - F(a) - \int_a^b f \right| < \varepsilon.$$

But, since $\varepsilon > 0$ is arbitrary, we infer that equation (1) holds. Q.E.D.

Remark If the function F is differentiable at every point of $[a, b]$, then (by Theorem 6.1.2) hypothesis (a) is automatically satisfied. If f is not defined for some point $c \in E$, we take $f(c) := 0$. Even if F is differentiable at every point of $[a, b]$, condition (c) is *not automatically satisfied*, since there exist functions F such that F' is not Riemann integrable. (See Example 7.3.2(e).)

7.3.2 Examples **(a)** If $F(x) := \frac{1}{2}x^2$ for all $x \in [a, b]$, then $F'(x) = x$ for all $x \in [a, b]$. Further, $f = F'$ is continuous so it is in $\mathcal{R}[a, b]$. Therefore the Fundamental Theorem (with $E = \emptyset$) implies that

$$\int_a^b x\, dx = F(b) - F(a) = \tfrac{1}{2}(b^2 - a^2).$$

(b) If $G(x) := \operatorname{Arctan} x$ for $x \in [a, b]$, then $G'(x) = 1/(x^2 + 1)$ for all $x \in [a, b]$; also G' is continuous, so it is in $\mathcal{R}[a, b]$. Therefore the Fundamental Theorem (with $E = \emptyset$) implies that

$$\int_a^b \frac{1}{x^2 + 1}\, dx = \operatorname{Arctan} b - \operatorname{Arctan} a.$$

(c) If $A(x) := |x|$ for $x \in [-10, 10]$, then $A'(x) = -1$ if $x \in [-10, 0)$ and $A'(x) = +1$ for $x \in (0, 10]$. Recalling the definition of the signum function (in 4.1.10(b)), we have $A'(x) = \operatorname{sgn}(x)$ for all $x \in [-10, 10] \backslash \{0\}$. Since the signum function is a step function, it belongs to $\mathcal{R}[-10, 10]$. Therefore the Fundamental Theorem (with $E = \{0\}$) implies that

$$\int_{-10}^{10} \operatorname{sgn}(x)\, dx = A(10) - A(-10) = 10 - 10 = 0.$$

(d) If $H(x) := 2\sqrt{x}$ for $x \in [0, b]$, then H is continuous on $[0, b]$ and $H'(x) = 1/\sqrt{x}$ for $x \in (0, b]$. Since $h := H'$ is not bounded on $(0, b]$, it does not belong to $\mathcal{R}[0, b]$ no matter how we define $h(0)$. Therefore, the Fundamental Theorem 7.3.1 does not apply. (However, we will see in Example 10.1.10(a) that h is *generalized* Riemann integrable on $[0, b]$.)

(e) Let $K(x) := x^2\cos(1/x^2)$ for $x \in (0, 1]$ and let $K(0) := 0$. It follows from the Product Rule 6.1.3(c) and the Chain Rule 6.1.6 that

$$K'(x) = 2x\cos(1/x^2) + (2/x)\sin(1/x^2) \qquad \text{for} \qquad x \in (0, 1].$$

Further, as in Example 6.1.7(e), it can be shown that $K'(0) = 0$. Thus K is continuous and differentiable at *every point* of $[0, 1]$. Since it can be seen that the function K' is not bounded on $[0, 1]$, it does not belong to $\mathcal{R}[0, 1]$ and the Fundamental Theorem 7.3.1 does not apply to K'. (However, we will see from Theorem 10.1.9 that K' is *generalized* Riemann integrable on $[0, 1]$.) □

The Fundamental Theorem (Second Form)

We now turn to the Fundamental Theorem (Second Form) in which we wish to differentiate an integral involving a variable upper limit.

7.3.3 Definition If $f \in \mathcal{R}[a, b]$, then the function defined by

$$F(z) := \int_a^z f \qquad \text{for} \qquad z \in [a, b], \tag{3}$$

is called the **indefinite integral** of f with **basepoint** a. (Sometimes a point other than a is used as a basepoint; see Exercise 6.)

We will first show that if $f \in \mathcal{R}[a, b]$, then its indefinite integral F satisfies a Lipschitz condition; hence F is continuous on $[a, b]$.

7.3.4 Theorem *The indefinite integral F defined by* (3) *is continuous on* $[a, b]$. *In fact, if* $|f(x)| \leq M$ *for all* $x \in [a, b]$, *then* $|F(z) - F(w)| \leq M|z - w|$ *for all* $z, w \in [a, b]$.

Proof. The Additivity Theorem 7.2.9 implies that if $z, w \in [a, b]$ and $w \leq z$, then

$$F(z) = \int_a^z f = \int_a^w f + \int_w^z f = F(w) + \int_w^z f,$$

whence we have

$$F(z) - F(w) = \int_w^z f.$$

Now if $-M \leq f(x) \leq M$ for all $x \in [a, b]$, then Theorem 7.1.5(c) implies that

$$-M(z - w) \leq \int_w^z f \leq M(z - w),$$

whence it follows that

$$|F(z) - F(w)| \leq \left|\int_w^z f\right| \leq M|z - w|,$$

as asserted. Q.E.D.

We will now show that the indefinite integral F is differentiable at any point where f is continuous.

7.3.5 Fundamental Theorem of Calculus (Second Form) *Let $f \in \mathcal{R}[a,b]$ and let f be continuous at a point $c \in [a,b]$. Then the indefinite integral, defined by* (3)*, is differentiable at c and $F'(c) = f(c)$.*

Proof. We will suppose that $c \in [a,b)$ and consider the right-hand derivative of F at c. Since f is continuous at c, given $\varepsilon > 0$ there exists $\eta_\varepsilon > 0$ such that if $c \leq x < c + \eta_\varepsilon$, then

$$f(c) - \varepsilon < f(x) < f(c) + \varepsilon. \tag{4}$$

Let h satisfy $0 < h < \eta_\varepsilon$. The Additivity Theorem 7.2.9 implies that f is integrable on the intervals $[a,c]$, $[a,\ c+h]$ and $[c,\ c+h]$ and that

$$F(c+h) - F(c) = \int_c^{c+h} f.$$

Now on the interval $[c,\ c+h]$ the function f satisfies inequality (4), so that we have

$$(f(c) - \varepsilon) \cdot h \leq F(c+h) - F(c) = \int_c^{c+h} f \leq (f(c) + \varepsilon) \cdot h.$$

If we divide by $h > 0$ and subtract $f(c)$, we obtain

$$\left| \frac{F(c+h) - F(c)}{h} - f(c) \right| \leq \varepsilon.$$

But, since $\varepsilon > 0$ is arbitrary, we conclude that the right-hand limit is given by

$$\lim_{x \to 0+} \frac{F(c+h) - F(c)}{h} = f(c).$$

It is proved in the same way that the left-hand limit of this difference quotient also equals $f(c)$ when $c \in (a,\ b]$, whence the assertion follows. Q.E.D.

If f is continuous on all of $[a,\ b]$, we obtain the following result.

7.3.6 Theorem *If f is continuous on $[a,\ b]$, then the indefinite integral F, defined by* (3)*, is differentiable on $[a,\ b]$ and $F'(x) = f(x)$* for all $x \in [a,b]$.

Theorem 7.3.6 can be summarized: *If f is continuous on $[a,\ b]$, then its indefinite integral is an antiderivative of f.* We will now see that, in general, the indefinite integral need not be an antiderivative (either because the derivative of the indefinite integral does not exist or does not equal $f(x)$).

7.3.7 Examples **(a)** If $f(x) := \operatorname{sgn} x$ on $[-1,\ 1]$, then $f \in \mathcal{R}[-1,\ 1]$ and has the indefinite integral $F(x) := |x| - 1$ with the basepoint -1. However, since $F'(0)$ does not exist, F is not an antiderivative of f on $[-1,\ 1]$.

(b) If h denotes Thomae's function, considered in 7.1.7, then its indefinite integral $H(x) := \int_0^x h$ is identically 0 on [0, 1]. Here, the derivative of this indefinite integral exists at every point and $H'(x) = 0$. But $H'(x) \neq h(x)$ whenever $x \in \mathbb{Q} \cap [0,\ 1]$, so that H is not an antiderivative of h on [0, 1]. □

Substitution Theorem

The next theorem provides the justification for the "change of variable" method that is often used to evaluate integrals. This theorem is employed (usually implicitly) in the evaluation by means of procedures that involve the manipulation of "differentials," common in elementary courses.

7.3.8 Substitution Theorem *Let $J := [\alpha, \beta]$ and let $\varphi : J \to \mathbb{R}$ have a continuous derivative on J. If $f : I \to \mathbb{R}$ is continuous on an interval I containing $\varphi(J)$, then*

$$\int_{\alpha}^{\beta} f(\varphi(t)) \cdot \varphi'(t)dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)dx. \tag{5}$$

The proof of this theorem is based on the Chain Rule 6.1.6, and will be outlined in Exercise 17. The hypotheses that f and φ' are continuous are restrictive, but are used to ensure the existence of the Riemann integral on the left side of (5).

7.3.9 Examples **(a)** Consider the integral $\displaystyle\int_1^4 \frac{\sin\sqrt{t}}{\sqrt{t}}dt$.

Here we substitute $\varphi(t) := \sqrt{t}$ for $t \in [1, 4]$ so that $\varphi'(t) = 1/(2\sqrt{t})$ is continuous on [1, 4]. If we let $f(x) := 2\sin x$, then the integrand has the form $(f \circ \varphi) \cdot \varphi'$ and the Substitution Theorem 7.3.8 implies that the integral equals $\int_1^2 2\sin x\,dx = -2\cos x|_1^2 = 2(\cos 1 - \cos 2)$.

(b) Consider the integral $\displaystyle\int_0^4 \frac{\sin\sqrt{t}}{\sqrt{t}}dt$.

Since $\varphi(t) := \sqrt{t}$ does not have a continuous derivative on [0, 4], the Substitution Theorem 7.3.8 is not applicable, at least with this substitution. (In fact, it is not obvious that this integral exists; however, we can apply Exercise 7.2.11 to obtain this conclusion. We could then apply the Fundamental Theorem 7.3.1 to $F(t) := -2\cos\sqrt{t}$ with $E := \{0\}$ to evaluate this integral.) □

We will give a more powerful Substitution Theorem for the *generalized* Riemann integral in Section 10.1.

Lebesgue's Integrability Criterion

We will now present a statement of the definitive theorem due to Henri Lebesgue (1875–1941) giving a necessary and sufficient condition for a function to be Riemann integrable, and will give some applications of this theorem. In order to state this result, we need to introduce the important notion of a null set.

Warning Some people use the term "null set" as a synonym for the terms "empty set" or "void set" referring to $\emptyset$ (= the set that has no elements). However, we will always use the term "null set" in conformity with our next definition, as is customary in the theory of integration.

7.3.10 Definition **(a)** A set $Z \subset \mathbb{R}$ is said to be a **null set** if for every $\varepsilon > 0$ there exists a countable collection $\{(a_k, b_k)\}_{k=1}^{\infty}$ of open intervals such that

$$Z \subseteq \bigcup_{k=1}^{\infty}(a_k, b_k) \qquad \text{and} \qquad \sum_{k=1}^{\infty}(b_k - a_k) \leq \varepsilon. \tag{6}$$

(b) If $Q(x)$ is a statement about the point $x \in I$, we say that $Q(x)$ holds **almost everywhere** on I (or for **almost every** $x \in I$), if there exists a null set $Z \subset I$ such that $Q(x)$ holds for all $x \in I \backslash Z$. In this case we may write

$$Q(x) \quad \text{for a.e.} \quad x \in I.$$

It is trivial that any subset of a null set is also a null set, and it is easy to see that the union of two null sets is a null set. We will now give an example that may be very surprising.

7.3.11 Example The $\mathbb{Q}_1$ of rational numbers in $[0, 1]$ is a null set.

We enumerate $\mathbb{Q}_1 = \{r_1, r_2, \ldots\}$. Given $\varepsilon > 0$, note that the open interval $J_1 := (r_1 - \varepsilon/4, r_1 + \varepsilon/4)$ contains r_1 and has length $\varepsilon/2$; also the open interval $J_2 := (r_2 - \varepsilon/8, r_2 + \varepsilon/8)$ contains r_2 and has length $\varepsilon/4$. In general, the open interval

$$J_k := \left(r_k - \frac{\varepsilon}{2^{k+1}}, r_k + \frac{\varepsilon}{2^{k+1}}\right)$$

contains the point r_k and has length $\varepsilon/2^k$. Therefore, the union $\bigcup_{k=1}^{\infty} J_k$ of these open intervals contains every point of $\mathbb{Q}_1$; moreover, the sum of the lengths is $\sum_{k=1}^{\infty} (\varepsilon/2^k) = \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $\mathbb{Q}_1$ is a null set. □

The argument just given can be modified to show that: *Every countable set is a null set.* However, it can be shown that there exist uncountable null sets in $\mathbb{R}$; for example, the Cantor set that will be introduced in Definition 11.1.10.

We now state Lebesgue's Integrability Criterion. It asserts that a bounded function on an interval is Riemann integrable if and only if its points of discontinuity form a null set.

7.3.12 Lebesgue's Integrability Criterion *A bounded function $f : [a, b] \to \mathbb{R}$ is Riemann integrable if and only if it is continuous almost everywhere on $[a, b]$.*

A proof of this result will be given in Appendix C. However, we will apply Lebesgue's Theorem here to some specific functions, and show that some of our previous results follow immediately from it. We shall also use this theorem to obtain the important Composition and Product Theorems.

7.3.13 Examples **(a)** The step function g in Example 7.1.4(b) is continuous at every point except the point $x = 1$. Therefore it follows from the Lebesgue Integrability Criterion that g is Riemann integrable.

In fact, since every step function has at most a finite set of points of discontinuity, then: *Every step function on $[a, b]$ is Riemann integrable.*

(b) Since it was seen in Theorem 5.6.4 that the set of points of discontinuity of a monotone function is countable, we conclude that: *Every monotone function on $[a, b]$ is Riemann integrable.*

(c) The function G in Example 7.1.4(d) is discontinuous precisely at the points $D := \{1, 1/2, \ldots, 1/n, \ldots\}$. Since this is a countable set, it is a null set and Lebesgue's Criterion implies that G is Riemann integrable.

(d) The Dirichlet function was shown in Example 7.2.2(b) not to be Riemann integrable.

Note that it is discontinuous at *every* point of $[0, 1]$. Since it can be shown that the interval $[0, 1]$ is not a null set, Lebesgue's Criterion yields the same conclusion.

(e) Let $h : [0, 1] \to \mathbb{R}$ be Thomae's function, defined in Examples 5.1.6(h) and 7.1.7.

In Example 5.1.6(h), we saw that h is continuous at every irrational number and is discontinuous at every rational number in $[0, 1]$. By Example 7.3.11, it is discontinuous on a null set, so Lebesgue's Criterion implies that Thomae's function is Riemann integrable on $[0, 1]$, as we saw in Example 7.1.7. □

We now obtain a result that will enable us to take other combinations of Riemann integrable functions.

7.3.14 Composition Theorem *Let $f \in \mathcal{R}[a, b]$ with $f([a, b]) \subseteq [c, d]$ and let $\varphi : [c, d] \to \mathbb{R}$ be continuous. Then the composition $\varphi \circ f$ belongs to $\mathcal{R}[a, b]$.*

Proof. If f is continuous at a point $u \in [a, b]$, then $\varphi \circ f$ is also continuous at u. Since the set D of points of discontinuity of f is a null set, it follows that the set $D_1 \subseteq D$ of points of discontinuity of $\varphi \circ f$ is also a null set. Therefore the composition $\varphi \circ f$ also belongs to $\mathcal{R}[a, b]$. Q.E.D.

It will be seen in Exercise 22 that the hypothesis that φ is continuous cannot be dropped. The next result is a corollary of the Composition Theorem.

7.3.15 Corollary *Suppose that $f \in \mathcal{R}[a, b]$. Then its absolute value $|f|$ is in $\mathcal{R}[a, b]$, and*

$$\left|\int_a^b f\right| \leq \int_a^b |f| \leq M(b - a),$$

where $|f(x)| \leq M$ for all $x \in [a, b]$.

Proof. We have seen in Theorem 7.1.6 that if f is integrable, then there exists M such that $|f(x)| \leq M$ for all $x \in [a, b]$. Let $\varphi(t) := |t|$ for $t \in [-M, M]$; then the Composition Theorem implies that $|f| = \varphi \circ f \in \mathcal{R}[a, b]$. The first inequality follows from the fact that $-|f| \leq f \leq |f|$ and 7.1.5(c), and the second from the fact that $|f(x)| \leq M$. Q.E.D.

7.3.16 The Product Theorem *If f and g belong to $\mathcal{R}[a, b]$, then the product fg belongs to $\mathcal{R}[a, b]$.*

Proof. If $\varphi(t) := t^2$ for $t \in [-M, M]$, it follows from the Composition Theorem that $f^2 = \varphi \circ f$ belongs to $\mathcal{R}[a, b]$. Similarly, $(f + g)^2$ and g^2 belong to $\mathcal{R}[a, b]$. But since we can write the product as

$$fg = \frac{1}{2}\left[(f + g)^2 - f^2 - g^2\right],$$

it follows that $fg \in \mathcal{R}[a, b]$. Q.E.D.

Integration by Parts

We will conclude this section with a rather general form of Integration by Parts for the Riemann integral, and Taylor's Theorem with the Remainder.

7.3.17 Integration by Parts *Let F, G be differentiable on $[a, b]$ and let $f := F'$ and $g := G'$ belong to $\mathcal{R}[a, b]$. Then*

$$\int_a^b fG = FG\Big|_a^b - \int_a^b Fg. \tag{7}$$

Proof. By Theorem 6.1.3(c), the derivative $(FG)'$ exists on $[a, b]$ and

$$(FG)' = F'G + FG' = fG + Fg.$$

Since F, G are continuous and f, g belong to $\mathcal{R}[a, b]$, the Product Theorem 7.3.16 implies that fG and Fg are integrable. Therefore the Fundamental Theorem 7.3.1 implies that

$$FG\Big|_a^b = \int_a^b (FG)' = \int_a^b fG + \int_a^b Fg,$$

from which (7) follows. Q.E.D.

A special, but useful, case of this theorem is when f and g are continuous on $[a, b]$ and F, G are their indefinite integrals $F(x) := \int_a^x f$ and $G(x) := \int_a^x g$.

We close this section with a version of Taylor's Theorem for the Riemann Integral.

7.3.18 Taylor's Theorem with the Remainder *Suppose that $f', \ldots, f^{(n)}, f^{(n+1)}$ exist on $[a, b]$ and that $f^{(n+1)} \in \mathcal{R}[a, b]$. Then we have*

$$f(b) = f(a) + \frac{f'(a)}{1!}(b-a) + \cdots + \frac{f^{(n)}(a)}{n!}(b-a)^n + R_n, \tag{8}$$

where the remainder is given by

$$R_n = \frac{1}{n!}\int_a^b f^{(n+1)}(t)\cdot(b-t)^n dt. \tag{9}$$

Proof. Apply Integration by Parts to equation (9), with $F(t) := f^{(n)}(t)$ and $G(t) := (b-t)^n/n!$, so that $g(t) = -(b-t)^{n-1}/(n-1)!$, to get

$$\begin{aligned} R_n &= \frac{1}{n!}f^{(n)}(t)\cdot(b-t)^n\Big|_{t=a}^{t=b} + \frac{1}{(n-1)!}\int_a^b f^{(n)}(t)\cdot(b-a)^{n-1}dt \\ &= -\frac{f^{(n)}(a)}{n!}\cdot(b-a)^n + \frac{1}{(n-1)!}\int_a^b f^{(n)}(t)\cdot(b-t)^{n-1}dt. \end{aligned}$$

If we continue to integrate by parts in this way, we obtain (8). Q.E.D.

Exercises for Section 7.3

1. Extend the proof of the Fundamental Theorem 7.3.1 to the case of an arbitrary finite set E.
2. If $n \in \mathbb{N}$ and $H_n(x) := x^{n+1}/(n+1)$ for $x \in [a, b]$, show that the Fundamental Theorem 7.3.1 implies that $\int_a^b x^n dx = \left(b^{n+1} - a^{n+1}\right)/(n+1)$. What is the set E here?
3. If $g(x) := x$ for $|x| \geq 1$ and $g(x) := -x$ for $|x| < 1$ and if $G(x) := \frac{1}{2}\left|x^2 - 1\right|$, show that $\int_{-2}^3 g(x)dx = G(3) - G(-2) = 5/2$.
4. Let $B(x) := -\frac{1}{2}x^2$ for $x < 0$ and $B(x) := \frac{1}{2}x^2$ for $x \geq 0$. Show that $\int_a^b |x|dx = B(b) - B(a)$.
5. Let $f : [a, b] \to \mathbb{R}$ and let $C \in \mathbb{R}$.
   (a) If $\Phi : [a, b] \to \mathbb{R}$ is an antiderivative of f on $[a, b]$, show that $\Phi_C(x) := \Phi(x) + C$ is also an antiderivative of f on $[a, b]$.
   (b) If Φ_1 and Φ_2 are antiderivatives of f on $[a, b]$, show that $\Phi_1 - \Phi_2$ is a constant function on $[a, b]$.

6. If $f \in \mathcal{R}[a,b]$ and if $c \in [a,b]$, the function defined by $F_c(z) := \int_c^z f$ for $z \in [a,b]$ is called the **indefinite integral** of f with **basepoint** c. Find a relation between F_a and F_c.

7. We have seen in Example 7.1.7 that Thomae's function is in $\mathcal{R}[0,1]$ with integral equal to 0. Can the Fundamental Theorem 7.3.1 be used to obtain this conclusion? Explain your answer.

8. Let $F(x)$ be defined for $x \geq 0$ by $F(x) := (n-1)x - (n-1)n/2$ for $x \in [n-1, n)$, $n \in \mathbb{N}$. Show that F is continuous and evaluate $F'(x)$ at points where this derivative exists. Use this result to evaluate $\int_a^b [\![x]\!]dx$ for $0 \leq a < b$, where $[\![x]\!]$ denotes the greatest integer in x, as defined in Exercise 5.1.4.

9. Let $f \in \mathcal{R}[a,b]$ and define $F(x) := \int_a^x f$ for $x \in [a,b]$.
   (a) Evaluate $G(x) := \int_c^x f$ in terms of F, where $c \in [a,b]$.
   (b) Evaluate $H(x) := \int_x^b f$ in terms of F.
   (c) Evaluate $S(x) := \int_x^{\sin x} f$ in terms of F.

10. Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a, b]$ and let $v : [c,d] \to \mathbb{R}$ be differentiable on $[c, d]$ with $v([c,d]) \subseteq [a,b]$. If we define $G(x) := \int_a^{v(x)} f$, show that $G'(x) = f(v(x)) \cdot v'(x)$ for all $x \in [c,d]$.

11. Find $F'(x)$ when F is defined on [0, 1] by:
    (a) $F(x) := \int_0^{x^2} (1+t^3)^{-1} dt$. (b) $F(x) := \int_{x^2}^x \sqrt{1+t^2}dt$.

12. Let $f : [0, 3] \to \mathbb{R}$ be defined by $f(x) := x$ for $0 \leq x < 1$, $f(x) := 1$ for $1 \leq x < 2$ and $f(x) := x$ for $2 \leq x \leq 3$. Obtain formulas for $F(x) := \int_0^x f$ and sketch the graphs of f and F. Where is F differentiable? Evaluate $F'(x)$ at all such points.

13. The function g is defined on [0, 3] by $g(x) := -1$ if $0 \leq x < 2$ and $g(x) := 1$ if $2 \leq x \leq 3$. Find the indefinite integral $G(x) = \int_0^x g$ for $0 \leq x \leq 3$, and sketch the graphs of g and G. Does $G'(x) = g(x)$ for all x in [0, 3]?

14. Show there does not exist a continuously differentiable function f on [0, 2] such that $f(0) = -1$, $f(2) = 4$, and $f'(x) \leq 2$ for $0 \leq x \leq 2$. (Apply the Fundamental Theorem.)

15. If $f : \mathbb{R} \to \mathbb{R}$ is continuous and $c > 0$, define $g : \mathbb{R} \to \mathbb{R}$ by $g(x) := \int_{x-c}^{x+c} f(t)dt$. Show that g is differentiable on $\mathbb{R}$ and find $g'(x)$.

16. If $f : [0, 1] \to \mathbb{R}$ is continuous and $\int_0^x f = \int_x^1 f$ for all $x \in [0, 1]$, show that $f(x) = 0$ for all $x \in [0, 1]$.

17. Use the following argument to prove the Substitution Theorem 7.3.8. Define $F(u) := \int_{\varphi(\alpha)}^u f(x)dx$ for $u \in I$, and $H(t) := F(\varphi(t))$ for $t \in J$. Show that $H'(t) = f(\varphi(t))\varphi'(t)$ for $t \in J$ and that

$$\int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)dx = F(\varphi(\beta)) = H(\beta) = \int_\alpha^\beta f(\varphi(t))\varphi'(t)dt.$$

18. Use the Substitution Theorem 7.3.8 to evaluate the following integrals.
    (a) $\int_0^1 t\sqrt{1+t^2}dt$, (b) $\int_0^2 t^2(1+t^3)^{-1/2}dt = 4/3$,
    (c) $\int_1^4 \frac{\sqrt{1+\sqrt{t}}}{\sqrt{t}}dt$, (d) $\int_1^4 \frac{\cos\sqrt{t}}{\sqrt{t}}dt = 2(\sin 2 - \sin 1)$.

19. Explain why Theorem 7.3.8 and/or Exercise 7.3.17 cannot be applied to evaluate the following integrals, using the indicated substitution.
    (a) $\int_0^4 \frac{\sqrt{t}dt}{1+\sqrt{t}}$ $\varphi(t) = \sqrt{t}$, (b) $\int_0^4 \frac{\cos\sqrt{t}\,dt}{\sqrt{t}}$ $\varphi(t) = \sqrt{t}$,
    (c) $\int_{-1}^1 \sqrt{1+2|t|}dt$ $\varphi(t) = |t|$, (d) $\int_0^1 \frac{dt}{\sqrt{1-t^2}}$ $\varphi(t) = \text{Arcsin } t$.